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Analysis of Autism Spectrum Disorder (ASD) Screening Data

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Abstract

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition that affects social communication and behavior. Early identification of ASD is important because timely intervention can significantly improve developmental outcomes for affected children. With the increasing availability of behavioral screening data, machine learning techniques have become a promising approach for supporting early ASD risk assessment. This study analyzes the AQ-Child dataset to develop predictive models that classify children as ASD-positive or ASD-negative based on behavioral questionnaire responses and demographic information.

The dataset contains 10,088 instances and 35 attributes, including 30 behavioral questionnaire items (Q1–Q30) and four demographic variables: age, gender, jaundice history, and family ASD history. Data preprocessing steps were performed to improve data quality, including handling missing values, removing duplicate records, and transforming categorical attributes into suitable formats for machine learning analysis. The classification models were implemented using the WEKA data mining platform and evaluated using 10-fold cross-validation.

Three supervised classification algorithms were applied: Decision Tree (J48), Random Forest, and Naïve Bayes. Model performance was assessed using accuracy, precision, recall, and F1-score derived from the confusion matrix. The experimental results show that the Random Forest classifier achieved the best performance with an accuracy of 82.9%, outperforming both the J48 Decision Tree (76%) and Naïve Bayes (82.1%) classifiers. The ensemble learning approach of Random Forest enables better handling of complex interactions among questionnaire attributes, resulting in improved predictive performance.

The findings demonstrate that machine learning techniques, particularly ensemble methods, can effectively analyze questionnaire-based screening data and support ASD risk prediction. Although the developed models are not intended for clinical diagnosis, they can assist in early screening and help identify children who may require further professional assessment. Future research may focus on feature selection, hyperparameter optimization, and the exploration of advanced ensemble models to further improve classification accuracy and generalization.

Table of Contents

Abstract.....	2
1. Introduction.....	4
2. Theory Background.....	5
2.1 Data Mining.....	5
2.1.1 Supervised and Unsupervised Learning.....	5
2.1.2 Classification Concept.....	6
2.1.3 Overfitting.....	6
2.1.4 Model Evaluation Basics.....	7
2.2 Analysis Methods.....	7
2.2.1 Decision Tree (J48).....	8
2.2.2 Random Forest Algorithm.....	9
2.2.3 Naïve Bayes.....	10
2.2.4 Test Object and Experimental Setup.....	11
2.3 Dataset.....	11
2.3.1 Dataset Overview.....	11
2.3.2 Dataset Source.....	12
2.3.3 Dataset Characteristics.....	12
2.3.4 Attribute Description.....	13
2.3.5 Target Variable.....	14
2.4 Data Preprocessing.....	14
2.4.1 Data Cleaning.....	14
2.4.2 Data Transformation.....	15
3. Analysis Results.....	16
3.1 Classification Result.....	16
3.1.1 Decision Tree (J48).....	16
3.1.2 Random Forest.....	17
3.1.3 Naïve Bayes.....	18
3.2 Discussion.....	19
4. Conclusion.....	20
References.....	21

List of Tables

Table	Description	Page
Table 2.1	Description of AQ-Child Dataset	13
Table 2.2	Demographic Attributes of AQ-Child Dataset	14
Table 2.3	Target Variable of AQ-Child Dataset	15
Table 3.1	Performance Results for J48 Decision Tree Classifier	18
Table 3.2	Confusion Matrix for J48 Classifier	18
Table 3.3	Performance Results for Random Forest	19
Table 3.4	Confusion Matrix for Random Forest	19
Table 3.5	Performance Results for Naïve Bayes Classifier	19
Table 3.6	Confusion Matrix for Naïve Bayes Classifier	20

1. Introduction

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition characterized by differences in social communication and interaction, along with restricted or repetitive behaviors. The prevalence of ASD has increased over the past decades due to improved awareness, broader diagnostic criteria, and better screening methods. According to the Diagnostic and Statistical Manual of Mental Disorders (DSM-5), ASD is considered a spectrum condition with varying levels of support needs rather than a single disorder [1]. Early detection is important because timely intervention can significantly improve developmental and social outcomes for individuals with ASD [2].

To support early identification, several screening tools have been developed. One widely used approach is questionnaire-based screening, such as the Autism Spectrum Quotient (AQ), which evaluates behavioral traits related to autism [3]. These screening methods are typically fast and inexpensive, making them suitable for large populations.

Recent research has explored the use of machine learning techniques to improve ASD screening analysis. Classification algorithms can learn patterns from screening datasets and predict whether an individual may have ASD based on questionnaire responses and demographic information. Previous studies have shown that models such as Decision Trees and Random Forests can achieve promising results when applied to ASD screening data [4].

In this assignment, an ASD screening dataset is analyzed in order to develop predictive models for classification. The dataset includes responses to screening questions together with background information such as age and gender. The task focuses on preparing the dataset and applying machine learning methods to classify individuals as likely ASD or non-ASD. Model performance is evaluated using metrics such as accuracy and confusion matrices.

The purpose of this analysis is not to provide medical diagnosis but to investigate how data-driven methods can support ASD screening. By studying the dataset and applying classification techniques, the report demonstrates how machine learning can assist in identifying patterns that may help improve early screening processes.

2. Theory Background

This chapter presents the theoretical foundation for the Autism Spectrum Disorder (ASD) risk analysis model based on the AQ-Child dataset. The objective is to analyze ASD risk using behavioral questionnaire responses (Q1–Q30) together with selected demographic attributes. Since the dataset contains labeled ASD outcomes, the task is formulated as a supervised binary classification problem. The structured nature of the AQ-Child data makes it suitable for machine learning-based classification. The proposed model is designed as a decision-support tool rather than a clinical diagnostic system.

This chapter is organized as follows. Section 2.1 introduces key concepts of data mining and supervised classification, including classification, overfitting, and model evaluation. Section 2.2 explains the classification algorithms used in this study: Decision Tree (J48), Random Forest, and Naïve Bayes. Section 2.3 describes the AQ-Child dataset, and Section 2.4 outlines the data preprocessing procedures. Overall, this chapter provides the theoretical background for the implementation and evaluation of the ASD prediction model.

2.1 Data Mining

Data mining is a systematic process of discovering meaningful patterns, correlations, and predictive knowledge from large datasets using computational and statistical techniques. It constitutes a central step in the Knowledge Discovery in Databases (KDD) framework and integrates principles from statistics, artificial intelligence, and machine learning. Data mining aims to extract valid, novel, and useful information from data in order to support decision-making and predictive analysis [5].

2.1.1 Supervised and Unsupervised Learning

Machine learning methods used in data mining are generally categorized as supervised or unsupervised learning, depending on whether labeled outcomes are available [6].

In supervised learning, each observation in the dataset is associated with a known target label. The objective is to learn a predictive model that maps input features to an output variable. The model is trained by minimizing prediction error on labeled data [6].

In contrast, unsupervised learning operates on data without predefined labels and aims to identify hidden patterns or group structures within the dataset [7].

Since the AQ-Child dataset includes a labeled outcome variable indicating ASD status (ASD = 0 or 1), the ASD risk prediction task is formulated as a supervised learning problem.

2.1.2 Classification Concept

Classification is a supervised learning technique in which the target variable is categorical [5]. In binary classification problems, the output belongs to one of two classes.

For the AQ-Child dataset, the two classes are:

- ASD = 1 (Positive)
- ASD = 0 (Negative)

The classifier learns a decision boundary that separates these classes based on demographic attributes and questionnaire responses. In probabilistic models, the classifier may also estimate the probability that a child belongs to the ASD-positive class [5]. Since the objective of this study is to estimate ASD risk and categorize children into risk groups, the task is defined as a binary classification problem.

2.1.3 Overfitting

Overfitting occurs when a predictive model captures noise in the training data instead of learning the true underlying patterns. As a result, the model performs well on training data but poorly on unseen data [7].

To prevent overfitting, it is essential to control model complexity and evaluate performance using independent test data or cross-validation techniques. In ASD risk prediction, avoiding overfitting is particularly important to ensure reliable screening performance.

2.1.4 Model Evaluation Basics

Model evaluation assesses how well a classifier performs on unseen data. For binary classification tasks, predictions are commonly summarized using a confusion matrix consisting of:

- True Positives (TP)
- True Negatives (TN)
- False Positives (FP)
- False Negatives (FN)

From these values, several evaluation metrics are derived:

- Accuracy measures overall correctness.
- Recall measures the proportion of actual ASD-positive cases correctly identified.
- Precision measures the reliability of positive predictions.
- F1-score provides a balanced measure of precision and recall.

In medical screening contexts such as ASD risk prediction, recall is particularly important because minimizing false negatives reduces the risk of missing children with ASD. Hence, the objective of this study is to prioritize high recall while maintaining balanced overall performance. Therefore, multiple evaluation metrics are considered rather than relying solely on accuracy.

Based on these theoretical principles of supervised classification and model evaluation discussed in Section 2.1, an ensemble learning approach is selected for implementation in the subsequent section.

2.2 Analysis Methods

In this study, a classification-based data mining approach is employed to analyze Autism Spectrum Disorder (ASD) risk using AQ-Child questionnaire responses and demographic attributes. Since the target variable is binary (ASD = 0 or 1), the task is formulated as a supervised classification problem.

The models are implemented using the WEKA data mining tool. Three classification algorithms are applied: Decision Tree (J48), Random Forest, and Naïve Bayes. These algorithms are selected because they represent different classification paradigms: decision

tree learning, ensemble learning, and probabilistic classification. Comparing these approaches allows evaluation of which method performs best for ASD screening data.

2.2.1 Decision Tree (J48)

Decision Tree (J48) is the implementation of the C4.5 algorithm developed by Quinlan [5]. It constructs a tree structure by recursively splitting the dataset into subsets based on attribute values.

In this study, attributes include questionnaire responses (Q1–Q30) and demographic variables such as age, gender, jaundice history, and family history of ASD. At each node, the algorithm selects the attribute that maximizes Information Gain or Gain Ratio. The dataset is partitioned until a stopping condition is met, such as when all instances in a node belong to the same class. After the full tree is built, pruning is applied to reduce overfitting and improve generalization [5].

The final model represents classification rules in a tree form, where internal nodes correspond to specific questionnaire items or demographic attributes, branches represent decision outcomes (e.g., Yes/No responses), and leaf nodes indicate the predicted ASD class.

The selection of the splitting attribute is based on Information Gain, which uses entropy as a measure of impurity:

$$Entropy(S) = - \sum_{i=1}^c p_i \log_2 p_i$$

where

S is the dataset,

p_i is the proportion of instances belonging to class i ,

c is the number of classes.

Information Gain is calculated as:

$$Gain(S, A) = Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$$

where

A is an attribute (e.g., Q5 or Gender),

S_v is the subset where attribute A takes value v .

2.2.2 Random Forest Algorithm

The Random Forest algorithm, proposed by Breiman [8], is an ensemble learning method based on decision trees. It combines multiple decision trees to improve predictive accuracy and generalization performance.

In this study, each tree is trained using different bootstrap samples drawn from the AQ-Child dataset. During tree construction, a random subset of features (from Q1–Q30 and demographic variables) is selected at each split. This randomness increases diversity among trees and reduces correlation between them [8].

For a classification task, the final prediction is obtained through majority voting across all trees:

$$\hat{y} = \text{mode}\{T_1(x), T_2(x), \dots, T_m(x)\}$$

where

$T_j(x)$ is the prediction of the j -th tree,

m is the total number of trees,

$\hat{y}$ is the final predicted class.

By averaging multiple decorrelated trees, Random Forest reduces variance while maintaining low bias, leading to improved stability compared to a single decision tree [7]. Because the AQ-Child dataset contains multiple behavioral indicators (Q1–Q30), ensemble learning helps capture complex interactions among questionnaire items.

2.2.3 Naïve Bayes

Naïve Bayes is a probabilistic classification algorithm based on Bayes' Theorem [9]. In this study, it predicts whether a child is ASD-positive or ASD-negative by computing the probability of each class given the observed questionnaire responses (Q1–Q30) and demographic features.

Bayes' Theorem is expressed as:

$$P(C_k | X) = \frac{P(X | C_k) P(C_k)}{P(X)}$$

where

C_k represents class k (ASD = 0 or 1),

$X = (x_1, x_2, \dots, x_n)$ represents the feature vector composed of Q1–Q30 responses and demographic attributes,

$P(C_k)$ is the prior probability of class C_k ,

$P(X | C_k)$ is the likelihood of observing features X given class C_k ,

$P(C_k | X)$ is the posterior probability.

Since $P(X)$ is constant for all classes, classification is performed by maximizing:

$$\hat{y} = \arg \max_{C_k} P(C_k)P(X | C_k)$$

To simplify computation, Naïve Bayes assumes that all features are conditionally independent given the class label. Therefore, the likelihood is computed as:

$$P(X | C_k) = \prod_{i=1}^n P(x_i | C_k)$$

In the context of this study, this means the model assumes that each questionnaire response (e.g., Q1, Q2, ..., Q30) and demographic attribute contributes independently to the probability of ASD. Although this independence assumption may not fully reflect real behavioral relationships, it allows efficient computation and often produces reliable classification results in questionnaire-based datasets [8].

2.2.4 Test Object and Experimental Setup

The test object in this study is the AQ-Child dataset, which contains 30 behavioral questionnaire items (Q1–Q30) along with demographic attributes such as age, gender, jaundice history, and family history of ASD. Each instance represents a child aged between 4 and 11 years.

All experiments are conducted using the WEKA data mining platform.

Classification experiments are performed using the following WEKA settings:

- Evaluation Method: 10-fold cross-validation
- Classifiers Used:
 - trees.J48
 - trees.RandomForest
 - bayes.NaiveBayes

Default parameter settings are applied unless otherwise specified:

- J48: Confidence Factor = 0.25, Minimum instances per leaf = 2
- Random Forest: Number of trees = 100, Random seed = 1
- Naïve Bayes: Default settings (no kernel estimator, no supervised discretization)

2.3 Dataset

2.3.1 Dataset Overview

The dataset used in this study is the AQ-Child Dataset, which is designed for early screening of Autism Spectrum Disorder (ASD) in children. The dataset is based on behavioral questionnaire responses and demographic information collected from children aged between 4 and 11 years.

The AQ-Child dataset is derived from the Autism Spectrum Quotient, which measures autistic traits in children through a structured set of behavioral questions. The questionnaire is typically completed by parents or guardians who observe the child's behavior in daily activities.

Analysis of Autism Spectrum Disorder (ASD) Screening Data

The primary objective of the dataset is to enable machine learning models to predict whether a child is likely to exhibit autism-related characteristics based on questionnaire responses and demographic information.

2.3.2 Dataset Source

The AQ-Child screening questionnaire was developed by NovoPsych, a digital mental health platform that provides psychological assessment tools used by healthcare practitioners, private clinics, and government health services.

The dataset contains responses collected from individuals who completed the questionnaire, and it has been widely used in machine learning research for ASD risk prediction.

2.3.3 Dataset Characteristics

The dataset contains 10,088 instances, where each instance represents one child's screening result. A total of 35 attributes are included in the dataset, consisting of demographic information, behavioral questionnaire responses, and the final ASD classification label.

Property	Description
Dataset Name	AQ-Child Dataset
Total Instances	10,088
Total Attributes	35
Behavioral Attributes	30 (Q1 – Q30)
Demographic Attributes	4
Missing Value	Present
Target Variable	ASD
Problem Type	Binary Classification

Table 2.1 Description of AQ-Child Dataset

2.3.4 Attribute Description

The dataset contains two main types of attributes: demographic attributes and behavioral questionnaire responses.

Demographic Attributes

These attributes describe general background information about the child.

Attribute	Description	Type
Age	Age of the child	Numeric
Gender	Gender of the child	Nominal
Jaundice	Whether the child had jaundice at birth	Nominal
Family_ASD_History	Whether there is ASD history in the family	Nominal

Table 2.2 Demographic Attributes of AQ-Child Dataset

These demographic variables may influence ASD risk prediction and therefore are included as input features for machine learning models.

Behavioral Questionnaire Attributes

The dataset includes **30 behavioral attributes (Q1 – Q30)** representing responses to the AQ-Child questionnaire. These questions measure behavioral patterns related to autism, such as:

- Social interaction
- Communication ability
- Attention switching
- Attention to detail
- Imagination

Each question is answered using a **Likert-scale response format**, which is converted into numerical values in the dataset to facilitate machine learning analysis.

2.3.5 Target Variable

The target variable in the dataset is **ASD**, which represents the screening result of the autism assessment.

Value	Meaning
1	ASD Positive (child shows autism-related traits)
0	ASD Negative (child shows no significant autism traits)

Table 2.3 Target Variable of AQ-Child Dataset

In the dataset, there are 4797 ASD-positive instances and 5291 ASD-negative instances, resulting in a relatively balanced class distribution. A balanced dataset is beneficial for supervised classification because it reduces the risk of biased model predictions toward the majority class.

2.4 Data Preprocessing

Data preprocessing is a crucial step in the data mining process. Raw datasets often contain inconsistencies, missing values, redundant entries, or noisy data that may negatively affect the performance of machine learning models. Proper preprocessing ensures that the dataset is clean, consistent, and suitable for analysis.

In this study, preprocessing techniques were applied to the AQ-Child Dataset before performing classification analysis using WEKA. These preprocessing steps helped improve data quality and ensured that the dataset could be effectively used for training and evaluating machine learning models.

2.4.1 Data Cleaning

Data cleaning is performed to identify and correct errors or inconsistencies in the dataset. The AQ-Child dataset contains missing values and duplicate entries that may influence model performance.

First, duplicate records are examined and removed in order to ensure that each instance represents a unique observation. Duplicate data can introduce bias in machine learning models because certain patterns may be over-represented during training.

Missing values are also addressed during preprocessing. Machine learning algorithms typically require complete data in order to perform correctly. In WEKA, missing values can be handled using built-in preprocessing filters. Numerical attributes may be replaced using the mean value of the attribute, while nominal attributes can be replaced with the most frequent category. This approach allows the dataset to remain usable without discarding a large number of instances.

In addition, the dataset is inspected for potential noise and outliers. For example, unrealistic values in numerical attributes such as age may indicate data entry errors. Detecting and examining such values helps ensure that the dataset remains consistent and reliable for analysis.

2.4.2 Data Transformation

Data transformation is performed to convert attribute values into a format that can be processed effectively by machine learning algorithms. In the AQ-Child dataset, most behavioral questionnaire responses (Q1–Q30) are already stored as numerical values representing the scoring of the Autism Spectrum Quotient (AQ) screening test. Therefore, these attributes do not require additional transformation before analysis.

However, several demographic attributes in the dataset are stored as categorical variables. These attributes include Gender, Jaundice, and Family_ASD_History. Since machine learning algorithms operate more efficiently with structured attribute representations, these categorical attributes are encoded as nominal values.

WEKA classifiers can directly process nominal attributes without requiring manual numerical conversion. As a result, these attributes are retained in nominal format while the numerical questionnaire responses remain unchanged.

This transformation ensures that all attributes are represented in a structured format suitable for classification algorithms while preserving the original meaning of the data.

Analysis of Autism Spectrum Disorder (ASD) Screening Data

The performance metrics obtained for the J48 classifier are summarized in Table 1.

Metric	Value(%)
Accuracy	76
Precision	76
Recall	76
F1-score	75.9

Table 3.1: Performance results for the J48 Decision Tree classifier.

The results indicate that the J48 classifier provides moderate predictive performance for ASD risk classification. Although the model correctly identifies many ASD and Non-ASD cases, the overall accuracy suggests that some misclassification still occurs.

The confusion matrix for the J48 classifier is presented in Table 2.

Actual/Predicted	ASD	Non-ASD
ASD	3062	1158
Non-ASD	959	3625

Table 3.2: Confusion matrix for J48 classifier.

The confusion matrix shows that the classifier correctly identified 3062 ASD-positive cases and 3625 ASD-negative cases. However, 2117 instances were misclassified, indicating the presence of both false positives and false negatives. This suggests that while the decision tree captures relevant patterns in the dataset, its classification performance remains limited compared with more advanced models.

3.1.2 Random Forest

The Random Forest classifier was applied to the same dataset to evaluate the effectiveness of ensemble learning for ASD risk prediction. The performance metrics obtained are summarized in Table 3.

Metric	Value(%)
Accuracy	82.9
Precision	83
Recall	82.9
F1-score	82.9

Table 3.3: Performance results for Random Forest.

The evaluation metrics indicate improved classification performance compared with the J48 model. The higher accuracy and balanced recall suggest that the Random Forest classifier is better able to capture patterns within the dataset. The confusion matrix for the Random Forest classifier is shown in Table 4.

Actual/Predicted	ASD	Non-ASD
ASD	3367	853
Non-ASD	651	3933

Table 3.4: Confusion matrix for Random Forest classifier.

The confusion matrix shows a higher number of correctly classified ASD and Non-ASD cases compared with the J48 model. Both false positives and false negatives are reduced, indicating improved predictive performance and more reliable classification results.

3.1.3 Naïve Bayes

In addition to tree-based models, the Naïve Bayes classifier was applied as a probabilistic baseline for comparison. The performance metrics obtained for the Naïve Bayes classifier are summarized in Table 5.

Metric	Value(%)
Accuracy	82.1
Precision	82.1
Recall	82.1
F1-score	82.1

Table 3.5: Performance results for Naïve Bayes classifier.

The results show that Naïve Bayes achieves performance comparable to the Random Forest model, with slightly lower overall accuracy. The metrics indicate that the classifier performs well in distinguishing between ASD and Non-ASD cases.

The confusion matrix for the Naïve Bayes classifier is presented in Table 6.

Actual/Predicted	ASD	Non-ASD
ASD	3371	849
Non-ASD	723	3861

Table 3.6: Confusion matrix for Naïve Bayes classifier.

The confusion matrix indicates that the classifier correctly identifies a large proportion of ASD and Non-ASD cases. However, some misclassification remains, particularly for ASD-negative instances. Overall, the model demonstrates competitive performance for questionnaire-based ASD classification.

3.2 Discussion

Based on the performance metrics, the Random Forest classifier achieved the highest overall accuracy (82.9%) and a balanced recall (82.9%), making it the most effective model for predicting ASD risk in this study. Its ensemble nature allows it to reduce variance and better generalize from the training data.

The J48 Decision Tree performed moderately, with an accuracy of 76%, but its misclassification rate was higher, especially for ASD-positive cases. Naïve Bayes achieved similar results to Random Forest but slightly underperformed in recall, likely due to the conditional independence assumption among questionnaire attributes.

All classifiers showed some misclassification of ASD-positive cases, highlighting that early screening datasets may contain variability, and subtle behavioral patterns that make prediction challenging. Despite preprocessing to remove missing values, duplicates, and outliers, these factors may still limit model performance. The analysis suggests that ensemble methods, such as Random Forest, are better suited for capturing complex interactions between behavioral and demographic attributes.

Potential improvements include feature selection or dimensionality reduction, alternative ensemble methods, and hyperparameter optimization to further improve predictive performance. Additional data collection could still help improve generalization, especially for underrepresented subgroups or rare patterns.

4. Conclusion

The analysis of the AQ-Child dataset was performed to predict Autism Spectrum Disorder (ASD) risk in children using machine learning classifiers. Among the three algorithms applied, J48 Decision Tree, Random Forest, and Naïve Bayes, the Random Forest classifier achieved the highest performance, with an accuracy of 82.9%, precision of 83%, recall of 82.9%, and F1-score of 82.9%. J48 provided moderate performance, while Naïve Bayes achieved comparable results with slightly lower recall. Misclassifications occurred due to inherent variability in behavioral responses and the complexity of the classification task.

The study demonstrates that ensemble methods are effective for capturing complex patterns in questionnaire-based ASD screening data. Future work may include hyperparameter optimization, additional feature engineering, exploration of advanced ensemble models, and exploration of advanced ensemble models to further enhance predictive accuracy and generalization.

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